

Cross-Species, Multimodal Pathology AI: A Comparative One Health Collaboration Plan

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1. Specific Aims (the collaboration anchors)

The premise is data-centric: AI becomes trustworthy in pathology only when expert-curated, verified, multimodal data drives it. The program is organized as one pipeline, **data curation** → **model and analysis** → **tooling and deployment**, with expert verification closing the loop. The PI leads the methods, the encoder, and the platform; collaborators provide the cases, comparative models, molecular and spatial data, clinical workflow, and validation. Each aim is a place where specific collaborations happen.

Central insight. Naturally occurring animal disease is the most efficient route to the data scale and diversity AI needs, and cross-species morphological conservation makes transfer principled rather than ad hoc. Collaboration is therefore the engine of the methods program, not a supplement to it: each partner supplies a slice of multimodal, multi-species data that no single lab holds, and the methods turn it into transferable, trustworthy models. The program's reach is set by the breadth of its partnerships; its independence, by the methods only the PI provides.

- **Aim 1 (data curation).** Assemble and register cross-species histology, spatial, and molecular data into an audited benchmark.
- **Aim 2 (model and analysis).** Build an open-weight cross-species encoder with interpretable, calibrated multimodal fusion.
- **Aim 3 (tooling and deployment).** Deploy an expert-in-the-loop tool whose corrections retrain it, validated in the diagnostic workflow.

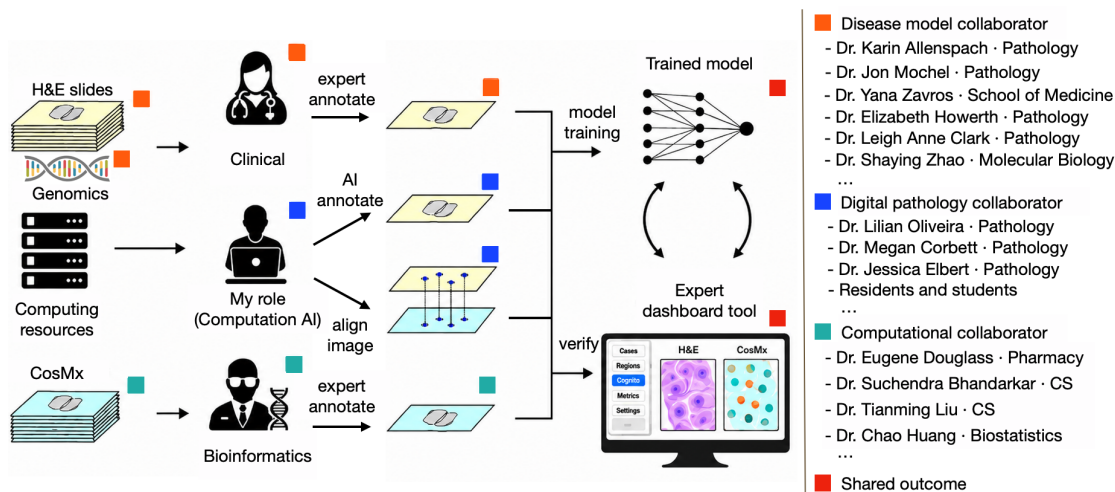


Figure 1. The collaboration model. Three groups of proposed partners (disease model, digital pathology, and computational) each contribute to one stage of a shared data pipeline, from expert-curated inputs (histology, genomics, spatial assays) through model training to an expert-verified tool. The PI contributes the computation and AI layer that connects the stages (annotation, registration, the encoder, the platform); collaborators contribute the cases, comparative models, molecular and spatial data, clinical workflow, and validation. The figure shows the proposed division of labor, not a claim of completed work.

2. How We Collaborate (shared mechanisms)

Every collaboration runs on the same lightweight mechanisms, so the program scales while keeping a clear

division of labor and an independent methods core:

- **Start small, then commit.** Each partnership opens with a single shared cohort (“look at one dataset together”), then expands to a funded aim once value is shown.
- **Clear division of labor.** The PI owns the methods, encoder, benchmarks, and platform; the partner owns the biology, the data, and the domain validation. This is stated per project, not assumed.
- **Shared credit.** Applied and methods contributions are recognized with authorship and roles agreed at the start of each project (ICMJE criteria).
- **Trainee involvement.** Where it fits, students and residents engage across pathology, computer science, and biostatistics, gaining both domain and computational exposure.
- **Shared platform instances.** Collaborators receive an instance of the no-code tool on their own data, so the collaboration produces something they use, not just a paper.
- **Joint funding.** PI-led methods grants (NSF AI core, CISE, CAREER) pair with collaborative application grants (NIH/NCI, including the Integrated Canine Data Commons; USDA/NIFA One Health).
- **Compounding data flywheel.** The expert-in-the-loop tool turns every correction into training signal, so each deployed instance improves the shared model. Partners join a system that gets better with use, not a one-off study.

3. Where We Collaborate (by partner)

The work sits across co-located units: the Department of Pathology and the Veterinary Teaching Hospital (cases, sign-out, deployment); the Precision One Health Initiative and its Systems Modeling and Data Analytics core; the School of Medicine (human translation); the AI institute and Computer Science (methods, students); Biochemistry/Bioinformatics and Biostatistics (molecular and statistical rigor); and institutional research computing (training). The partnerships below are proposed; each reflects a natural scientific complementarity and would be developed together with the collaborator.

(a) Disease-model partners: cases, comparative models, molecular ground truth (Aim 1, validation)

- **Dr. Karin Allenspach** (Pathology; lead, POHI Translational Medicine Core). *Where:* patient-derived canine organoids and the Canine Bladder Cancer Consortium, at curation (Aim 1). *How:* Dr. Allenspach contributes organoid cultures and matched histology; the PI images organoids and registers morphology to molecular readouts, producing a shared organoid-imaging resource. *Why it matters:* Organoids are the one setting where the same tissue is both imaged and perturbed, closing the morphology-to-molecular-to-response loop.
- **Dr. Jon Mochel** (Pathology; founding Director, Precision One Health Initiative). *Where:* POHI’s One Health translational program and quantitative systems-pharmacology modeling. *How:* the PI’s imaging and multimodal methods add the observational tissue layer the modeling otherwise lacks; together we develop the multimodal patient representation, with regular meetings and potential POHI pilot support. *Why it matters:* Imaging grounds population and systems models in the individual’s tissue state, the observational layer a digital-twin direction otherwise lacks.
- **Dr. Yana Zavros** (GRA Eminent Scholar, School of Medicine). *Where:* human-side comparator cohorts and patient-derived models, the human-translation step (Aims 1 and 3). *How:* Dr. Zavros contributes de-identified human comparator cases; together we validate canine-to-human transfer on shared diseases (e.g. endocrine tumors), and Dr. Zavros leads the human-translation evaluation of the tool. *Why it*

matters: The human side closes the One Health loop, so a model proven on abundant animal disease earns a credible path to the rarer human disease.

- **Dr. Elizabeth Howerth** (Pathology; anatomic-pathology service). *Where*: the diagnostic ground-truth standard, at annotation (Aim 1) and validation (Aim 3). *How*: Dr. Howerth adjudicates hard cases and sets the annotation standard; sign-out corrections become validation labels; Dr. Howerth defines what “trustworthy output” must clear. *Why it matters*: Her corrections are the calibration signal that makes the model’s abstention trustworthy, not only its predictions.
- **Dr. Leigh Anne Clark** (Pathology; canine genetics). *Where*: canine GWAS cohorts with germline and phenotype data, at curation (Aim 1) and molecular features (Aim 2). *How*: Dr. Clark contributes paired genotype and phenotype data for cases with matched imaging; the PI integrates genotype with morphology; output is a joint genotype-to-morphology study. *Why it matters*: Linking genotype to tissue makes heritable risk visible as morphology, a phenotype bridge sequence alone cannot draw.
- **Dr. Shaying Zhao** (Biochemistry and Molecular Biology / Bioinformatics). *Where*: a large canine cancer genome cohort and mutation-discovery pipelines, the molecular half of Aim 2. *How*: Dr. Zhao contributes somatic-mutation and driver calls; the PI supplies the imaging encoder and registration to map morphology to drivers; the model fuses both. *Why it matters*: Her cohorts are sequencing-only; imaging reveals which morphology a driver mutation actually produces, spatially grounding the dog and human comparison.

(b) Digital-pathology partners: workflow, spatial data, deployment (Aims 2–3)

- **Dr. Lilian Oliveira** (Assistant Professor, Pathology; spatial biology; AI-institute affiliate). *Where*: spatial-transcriptomics and molecular-pathology data, at spatial registration (Aim 1–2) and biomarker fusion (Aim 2); a primary scientific overlap. *How*: Dr. Oliveira runs spatial assays; the PI registers spots to H&E and builds interpretable fusion with pathway co-attention; we co-lead a spatial-omics project and connect the program to the AI institute. *Why it matters*: Registration turns spatial spots from “near this region” to “on this exact morphology,” making biomarker discovery reproducible rather than eyeballed.
- **Dr. Megan Corbett** (Pathology; anatomic pathology, digital pathology and AI). *Where*: digital-pathology and AI expertise plus translational organoid models, at deployment and validation (Aim 3). *How*: Dr. Corbett helps scope and test the tool in the anatomic-pathology service; expert corrections retrain the model and we measure reproducibility and turnaround. *Why it matters*: An AI-fluent anatomic pathologist can both validate the model and co-design the workflow it lives in, a rare combination.
- **Dr. Jessica Elbert** (Pathology; diagnostic and wildlife pathology). *Where*: diverse cross-species and wildlife cases, at curation (Aim 1) and second-reader validation (Aim 3). *How*: Dr. Elbert contributes species breadth and a second-read use case to test reproducibility where diagnosis is slowest or least consistent. *Why it matters*: Wildlife and non-domestic cases are the hardest distribution shift, the toughest test of whether cross-species transfer is real.
- **Residents and students** (pathology training program). *Where*: annotation throughput and the next-generation user base, across Aims 1 and 3 and teaching/service. *How*: course- and rotation-based annotation tasks feed curation; trainees use and stress-test the tool, linking the program to a teaching line. *Why it matters*: Trainees who both annotate and use the tool make it self-sustaining and surface failure modes a finished system would hide.

(c) Computational partners: methods depth, scale, statistical rigor (Aim 2)

- **Dr. Eugene Douglass** (Pharmaceutical and Biomedical Sciences, College of Pharmacy; lead, POHI Systems Modeling and Data Analytics Core). *Where:* the Systems Modeling and Data Analytics Core, and data-science and AI for drug-response and treatment matching including the PANGEA drug-perturbation RNAseq database (Aim 2). *How:* the PI's imaging and multimodal methods plug into the core's modeling and analytics work; PANGEA enters as a longer-term molecular data source. *Why it matters:* The Systems Modeling and Data Analytics Core is the natural home for the imaging and AI layer within POHI, and linking drug-response data points the program toward therapy prediction.
- **Dr. Suchendra Bhandarkar** (Computer Science). *Where:* computer-vision methods and a CS/AI-institute student pipeline (Aim 2). *How:* collaborate on the encoder's vision methods. *Why it matters:* Computer-vision depth lets a pathology-specific encoder build on current methods rather than reinvent them.
- **Dr. Tianming Liu** (Computer Science; foundation models). *Where:* foundation-model pretraining expertise (Aim 2). *How:* co-develop a pretraining recipe for veterinary tissue and share pretraining know-how toward an open-weight encoder release. *Why it matters:* A veterinary-tissue pretraining recipe is what makes a cross-species encoder more than a fine-tuned human model.
- **Dr. Chao Huang** (Biostatistics, College of Public Health). *Where:* statistical learning for imaging genetics and high-dimensional, functional, and manifold biomedical data (Aim 2). *How:* Dr. Huang contributes statistical rigor (calibration and imaging-genetics methods); we compare statistical and learned approaches on shared data, a natural early-stage methods collaboration. *Why it matters:* Calibration and inference are what let the model's confidence be believed, the precondition for clinical trust.

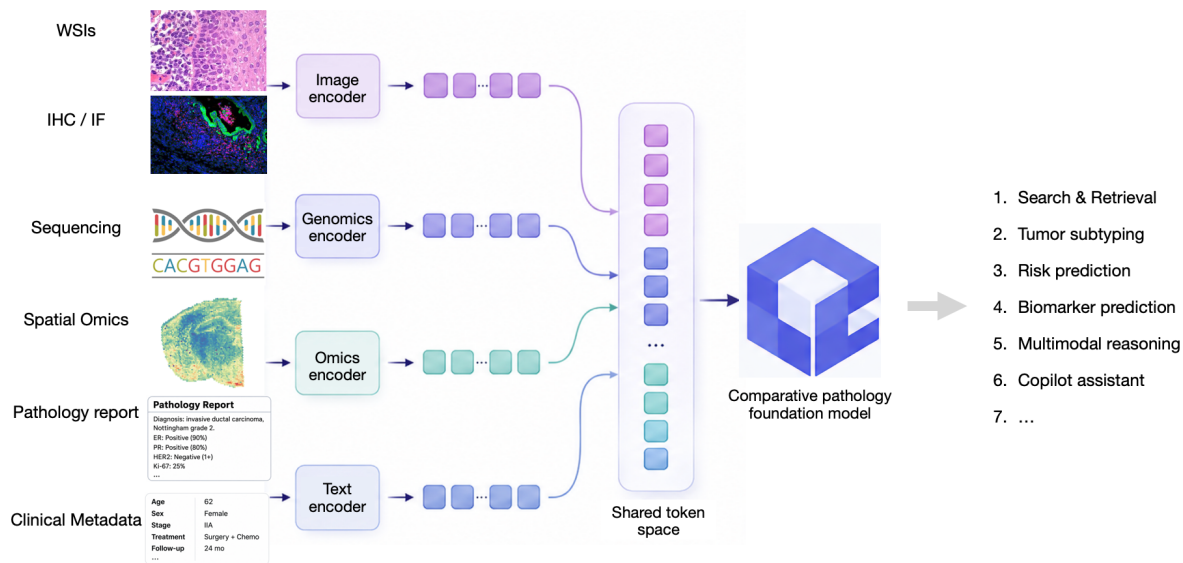


Figure 2. Aim 2, the comparative pathology foundation model. Modality-specific encoders map whole-slide images, IHC/IF, sequencing, spatial omics, pathology reports, and clinical metadata into a shared token space, supporting retrieval, tumor subtyping, risk and biomarker prediction, multimodal reasoning, and a copilot assistant.

4. Coordination and Governance

- **Decision-making.** The PI directs methods, platform, and benchmarks and coordinates overall direction; each partner directs the science and validation in their domain.

- **Communication.** A recurring joint meeting across the three groups, with pairwise project meetings as aims launch and an annual milestone-and-authorship review.
- **Data and compliance.** Shared governance under existing IACUC/IRB protocols and data-use agreements; curated datasets, benchmarks, and open model weights released where consent and policy allow, consistent with the NIH Data Management and Sharing Policy.
- **Conflict resolution.** Resolved by the relevant domain lead and the PI together; unresolved issues escalate to department or initiative leadership.

5. Positioning and Risk

Most computational-pathology efforts sit at one pole: foundation-model groups that build encoders but never deploy, or clinical-AI groups that deploy narrow tools without cross-species methods. This program occupies the rare middle: methods-led and deployed, and cross-species by design. That middle is also where the comparative One Health data is uniquely available. The chief collaboration risk is the opposite failure, over-collaboration that dissolves into a service core. The explicit guard is the independent methods spine (the driving question, the encoder, the platform, and PI-led funding), which stays with the PI regardless of how many partners join. The main practical risk, timely access to curated multimodal cohorts, is mitigated by the start-small handshakes and the shared platform instances that make early participation low-cost and immediately useful to partners.

6. Outputs

Each collaboration produces a usable artifact for the partner (a dataset, a registered cohort, or a platform instance on their data) alongside shared scholarly output. Across partners these compound into an open-weight cross-species model, a multimodal benchmark, and a deployed expert-in-the-loop tool (the OnePath copilot, Figure 3), advancing veterinary care and human translational research under one program. Because the partners share one model, one benchmark, and one platform, the collaborations compound: each curated cohort raises the benchmark for all, each registration method transfers to the next modality, and each deployed instance feeds the flywheel. The program's value is meant to grow faster than the sum of its partnerships.

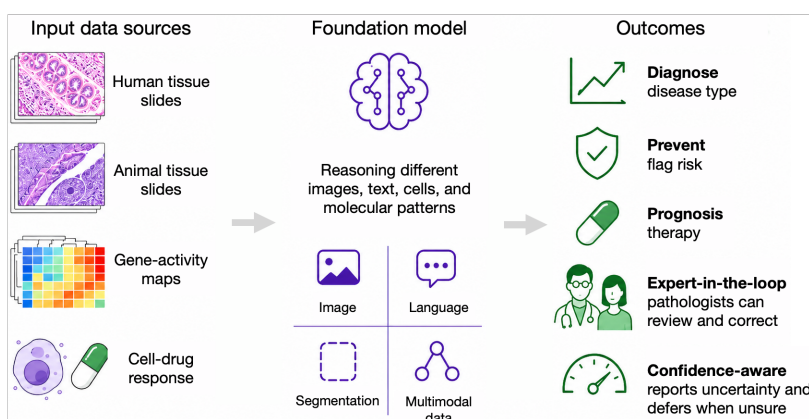


Figure 3. The OnePath copilot (Aim 3), the deployed expert-in-the-loop product. Human and animal slides, gene-activity maps, and cell-drug response feed a foundation model whose outputs pathologists review and correct, and which is confidence-aware, deferring when uncertain.